CO-1 AT-3

Name: S. Divya

MLA0303 - Reinforcement learning

Reg No: 192425163

Slot - A

① Develop a python-based Reinforcement learning model using OpenAI Gym to solve a Grid World navigation problem. Explain the state space action space, reward function, and constraints.

Introduction:

Reinforcement learning (RL) is a type of machine learning where an agent learns by interacting with an environment. The agent receives rewards for good actions and penalties for bad actions. In this problem, the agent moves in a grid world and learns the shortest safe path to reach the goal while avoiding obstacles.

Working:

i) Agent starts from the initial position.

ii) Chooses an action.

iii) Receives reward.

iv) Updates Q- values.

v) Repeats until the goal is reached

Advantages:

* learns optimal path automatically.
* Avoids obstacles.
* Improves performance after every episode

② Implement an ε-Greedy Multi-Armed Bandit algorithm and compare $\varepsilon = 0.1, 0.3$, and 0.5.

Introduction:

The Multi-Armed Bandit problem helps an agent decide whether to explore new actions or exploit the best-known action. The ε-Greedy algorithm balances exploration and exploitation.

Working Principle:

* With probability ε, the agent explores randomly.
* With probability $1-\varepsilon$, the agent chooses the best action.

Advantages:

* Simple to implement
* prevents the agent from getting stuck
* Finds better actions over time.

(3) Design a Markov Decision process (MDP) for an autonomous robot minimizing energy consump-tion.

Introduction:

A Markov Decision process (MDP) is a mathematical model used to solve sequential decision making problems. The robot chooses actions that minimize energy while reaching the destination.

Components of MDP:

* States
* Actions
* Transition Probability
* Energy Constraints
* Reward Function

Working:

i) Observe current state

ii) Select action

iii) Move to next state

iv) Receive reward.

v) Continue until destination.

Advantages:

* Reduces battery consumption
* Finds energy-efficient routes
* Makes intelligent decisions.

(4) Implement Value Iteration for a constrained Grid Environment. Explain Bellman Equation.

Introduction:

Value Iteration is a Reinforcement learning algorithm that calculates the best value for every state. It uses the Bellman equation repeatedly until the values become stable.

Working:

i) Initialize all state values to zero.

ii) Update values using Bellman Equation.

iii) Repeat until values stop changing.

iv) Select the action with the highest value

Advantages:

* Finds optimal policy
* Handles restricted environments
* Improves navigation efficiency.

(5) A warehouse robot has a battery capacity of only 30 minutes Design a RL framework to maximize deliveries without exceeding the battery limit.

Introduction:

A warehouse robot must complete as many deliveries as possible before its battery runs out. Reinforcement learning helps robot learn the most efficient delivery strategy.

Working:

i) Observe battery level.

ii) Select the next delivery.

iii) Calculate shortest route

iv) Deliver package

v) Return for charging when needed

Advantages:

* Increases delivery efficiency

* Saves battery power.

* Reduces travel time.

* Improves warehouse productivity.